Affinity Drifting: Learning Conditional Distributions from One Sample per Condition

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Working draft — results update automatically as experiments finish (2026-11-08 15:19 UTC, commit `9952a88`)

Abstract

Drifting models train a one-step generator by moving its samples along a field $V_{p,q} = V_p^+ - V_q^-$ that attracts them toward positive samples from the data distribution p and repels them from negative samples of the generated distribution q. In text-to-image data, however, each caption usually comes with a single image, so the empirical conditional $p_{\text{data}}(\cdot \mid c)$ is a point mass. We show that the attraction toward a point mass reduces exactly to ℓ_2 regression, so a generator that cannot resolve nearby conditions collapses to the conditional mean. We propose Affinity Drifting, which draws positives from a Nadaraya–Watson estimate of the conditional distribution, pooling the samples of semantically similar conditions with weights from a fixed condition embedding, while keeping the sample-space kernel of drifting. For Gaussian kernels the field is a scaled difference of smoothed log-densities, which identifies its fixed points: with negatives from the condition's own generated samples the equilibrium is the Nadaraya–Watson mixture, while pooling the negatives in the same way memorizes the training positives. We test this with pre-registered, one-factor-at-a-time experiments. In a non-parametric particle system the predicted fixed points are reached at the resolution of the kernel. In a 2-D toy with known conditional distributions, the original Drifting objective and the one-positive objective both collapse to the conditional mean (ratio of total variances $\operatorname{tr}\Sigma_q / \operatorname{tr}\Sigma_p$ of 2.8×10^{-5} and 3.8×10^{-4}), while an oracle with many samples per condition does not (0.909). Affinity Drifting restores the mixture: it captures every mode and lowers the held-out Wasserstein-2 distance to 0.33 of the one-positive objective, on par with the oracle. Ablations show why: pooling the positives without the sample-space kernel still collapses, and random instead of semantically similar neighbors place samples off the conditional support.

1. Introduction

Drifting models [2] train a one-step generator f_θ by evolving its pushforward distribution $q_\theta = (f_\theta)_\# p_\epsilon$ during training rather than at sampling time: each generated sample is attracted toward positive samples $y^+ \sim p$ and repelled from negative samples $y^- \sim q_\theta$, and the generator regresses onto its own drifted outputs. With class labels every class has many images, and the attraction is a kernel mean-shift toward the class-conditional distribution $p_{\text{data}}(\cdot \mid c)$.

Text conditioning breaks this assumption. A caption typically describes one image, so the empirical $p_{\text{data}}(\cdot \mid c)$ is a single point mass. The normalized kernel weight of that one positive is then identically one, the attraction no longer depends on where the sample is, and drifting reduces to ℓ_2 regression onto the positive (Lemma 1). A generator of finite resolution in c averages the positives of the conditions it cannot distinguish and outputs their mean, a blurred sample between modes (Fig. 1).

We ask whether pooling the positives of semantically similar conditions restores a distribution, and why. Affinity Drifting replaces the point mass by a Nadaraya–Watson estimate of $p_{\text{data}}(\cdot \mid c)$ and keeps the sample-space kernel $k(x, y^+)$ of drifting, so the attraction becomes a mean-shift toward a mixture rather than a pull toward an average. The claim is mechanistic: the gain should come from the x-dependent assignment of samples to modes, not merely from having more targets. We test it with ablations that remove one alternative explanation at a time: pooling positives with a constant sample-space kernel, pooling random rather than similar neighbors, and perturbing the single positive with matched noise, against an oracle with true conditional samples and the original Drifting objective as a baseline.

Contributions. (i) The one-positive degeneracy of drifting and its consequence, mean collapse under limited conditional capacity. (ii) Affinity Drifting, with a characterization of its fixed points for Gaussian kernels. (iii) A pre-registered, one-factor-at-a-time evaluation, run from a non-parametric particle system to a neural 2-D toy. This draft reports finished experiments and marks the others as pending.

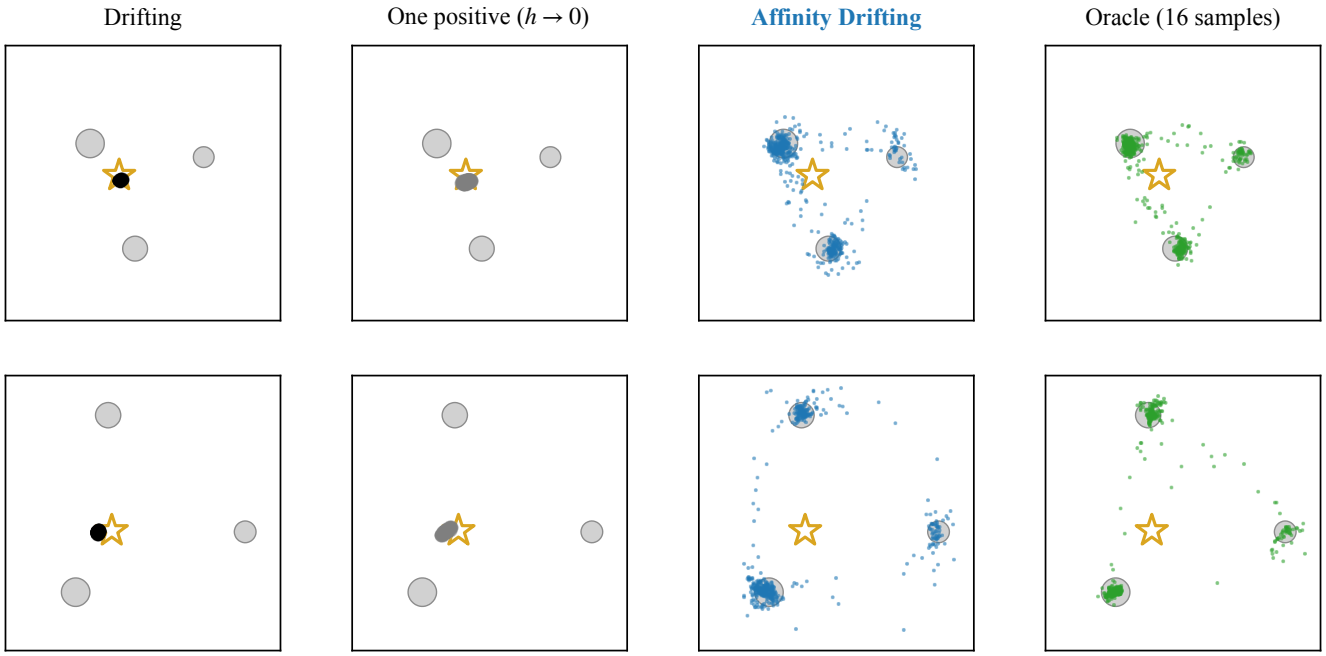


Figure 1. **Mean collapse with one sample per condition.** Samples (512 per panel) from generators trained on a 2-D toy in which each training condition c has exactly one sample, shown for two held-out conditions. Grey: the true $p_{\text{data}}(\cdot \mid c)$, a three-component Gaussian mixture; hollow $\star$: the conditional mean $\mathbb{E}[x \mid c]$. The original Drifting objective and the one-positive objective place every sample at the conditional mean, between the modes; an oracle given many true samples per condition covers the modes. Affinity Drifting uses only the single training samples, pooled across similar conditions, and recovers the modes. Seed 0 of five.

2. Method

Drifting. A generator $f_\theta : \mathbb{R}^C \times \mathcal{C} \to \mathbb{R}^D$ maps noise $\epsilon \sim p_\epsilon$ and a condition c to a sample $x = f_\theta(\epsilon, c)$, with conditional pushforward $q_\theta(\cdot \mid c)$. Following [2], the drifting field is

$$V_p^+(x) = \frac{1}{Z_p(x)} \mathbb{E}_p\big[k(x, y^+)(y^+ - x)\big], \qquad (1)$$

with $Z_p(x) = \mathbb{E}_p[k(x, y^+)]$, V_q^- the same expression over $y^- \sim q$, and $V_{p,q} = V_p^+ - V_q^-$. With $p = p_{\text{data}}(\cdot \mid c)$ and $q = q_\theta(\cdot \mid c)$ the generator is trained with

$$\mathcal{L} = \mathbb{E}_{c,\epsilon}\big\| f_\theta(\epsilon, c) - \text{stopgrad}\big(f_\theta(\epsilon, c) + V_{p,q_\theta}(f_\theta(\epsilon, c))\big)\big\|^2. \qquad (2)$$

In practice V is replaced by the normalized drift $\tilde{V} = V/\lambda$, $\lambda = \big(\mathbb{E}[\frac{1}{D}\|V\|^2]\big)^{1/2}$ over the batch, summed over several kernel scales [2].

Let the training set be $\{(c_i, y_i)\}_{i=1}^N$ with distinct conditions, so that the empirical conditional at c_i is δ_{y_i}.

Lemma 1 (One positive is regression) *If $p = \delta_y$, then $V_p^+(x) = y - x$ for every kernel $k > 0$. Without negatives, the gradient of (2) equals that of $\mathbb{E}_\epsilon\|f_\theta(\epsilon, c) - y\|^2$.*

Affinity Drifting. We draw positives from the Nadaraya–Watson estimate of the conditional distribution [8, 10],

$$\hat{p}_h(\cdot \mid c) = \sum_{i=1}^N \tilde{w}_i(c)\, \delta_{y_i}, \qquad \tilde{w}_i(c) = \frac{w_i(c)}{\sum_{j=1}^N w_j(c)}, \qquad (3)$$

where $w_i(c) = \exp\big(-\|\psi(c) - \psi(c_i)\|^2/2h^2\big)$ with a fixed condition embedding ψ (the true coordinates in the toy; a frozen text encoder in later stages), and keep (1) with $p = \hat{p}_h(\cdot \mid c)$. Negatives are drawn from the analogous estimate $\hat{q}_{h'}(\cdot \mid c) = \sum_i \tilde{w}_i(c)\, q_\theta(\cdot \mid c_i)$ with its own bandwidth h'; by default $h' = 0$, *i.e.* $q_\theta(\cdot \mid c)$ itself. The bandwidth $h = h(c)$ is set per condition to the smallest value whose effective sample size [4] $n_{\text{eff}}(c) = (\sum_i w_i)^2/\sum_i w_i^2$ reaches $n^\star = 16$, so every condition pools equally many positives. The limit $h \to 0$ recovers the one-positive objective, and $h, h' \to \infty$ unconditional drifting.

Fixed points. For a Gaussian kernel $k(x, y) = \exp(-\|x - y\|^2/2\sigma^2)$, Z_p is proportional to the density of $p * \mathcal{N}(0, \sigma^2 I)$, and $\nabla_x Z_p(x) = \sigma^{-2}\, \mathbb{E}_p[k(x, y^+)(y^+ - x)]$. Hence:

Lemma 2 (Score difference) *For the Gaussian kernel,*

$$V_{p,q}(x) = \sigma^2\, \nabla_x \big[\log Z_p(x) - \log Z_q(x)\big].$$

$V_{p,q} \equiv 0$ therefore holds if and only if $p * \mathcal{N}(0, \sigma^2 I) = q * \mathcal{N}(0, \sigma^2 I)$, *i.e.* $q = p$, since Gaussian convolution is injective on probability measures. With $h' = 0$ the equilibrium of each condition is the Nadaraya–Watson mixture, $q_\theta(\cdot \mid c) = \hat{p}_h(\cdot \mid c)$. With $h' = h$ the weights $\tilde{w}_i(c)$ act on both sides, and $q_\theta(\cdot \mid c_i) = \delta_{y_i}$, memorization of the training positives, is a fixed point for every c. The negative bandwidth h' thus selects the equilibrium.

Implementation. We use three Gaussian scales $\sigma \in \{0.071, 0.384, 2.07\}$, from the $n^\star$-th-nearest-neighbor distance of the positives to the median pairwise distance (the median heuristic [3]), and sum one normalized field per scale as in the multi-temperature drift of [2]; each scale satisfies Lemma 2, so the fixed point is unchanged. As in [2], the normalization λ is computed for the whole field $V_{p,q}$, not separately for V_p^+ and V_q^-. Separate normalization, which we used first, keeps both terms at unit scale even where they should cancel, and sent a few percent of one-positive samples far outside the data (Sec. 4). Drifting uses the kernel $\exp(-\|x - y\|/\tau)$; we use the Gaussian kernel so that Lemma 2 applies, and report the original kernel and normalization unchanged as the baseline.

3. Experiments

Protocol. Each comparison changes a single factor of a fixed base configuration; a run does not start if its configuration differs from the declared change. Predictions, falsification criteria and decision rules are committed before any run, and every outcome (pass, fail or inconclusive) is computed from those rules only. All methods share the seeds (five per experiment), the evaluation conditions and the learning rate, which is selected once on an unconditional model. Intervals are 95% paired bootstrap confidence intervals over seeds (10^4 resamples). Every experiment also trains the original Drifting objective, unchanged, as a reference that is reported but not judged.

3.1. Fixed points in a particle system

Without a network, each of 32 conditions on a line holds 256 particles updated by $x \leftarrow x + \eta V(x)$, a generator of unlimited capacity, with one positive per condition. The field agrees with the automatic-differentiation score difference of Lemma 2 to 1.6×10^{-15} in float64 (PASS). Increasing h' from 0 to h moves the equilibrium monotonically from the Nadaraya–Watson mixture to memorization (Fig. 2; PASS), whereas the original Drifting objective memorizes outright (share of particles nearest their own positive: 0.951).

The assignment of particles to individual positives, however, misses the pre-registered tolerance for the mixture (total variation 0.147 [0.145, 0.149] against 0.1; FAIL): with a kernel wider than the spacing of the positives, the field cannot resolve which nearby positive a particle sits on. Compared at the resolution of the kernel — total variation between the σ-smoothed densities, an analysis that was not pre-registered — the equilibrium matches $\hat{p}_h$ to 1.0×10^{-4}. We also draw negatives from another condition's data with weight λ, a two-condition analogue of the guidance in [2]. For Gaussian conditionals the predicted equilibrium is Gaussian in closed form, and the particle system reaches it under a Gaussian moment closure (relative error 3.3×10^{-6}). With finite samples, however, it is stable only for small weights (error 0.0278 at $\lambda = 0.25$, divergence from $\lambda = 0.5$; FAIL).

Method	$W_2 \downarrow$	$\operatorname{tr}\Sigma_q / \operatorname{tr}\Sigma_p$	Prec. ↑	Rec. ↑	Modes ↑
Drifting	1.4	2.8×10^{-5}	0.0023	0	0.34
One positive ($h \to 0$)	1.4	3.8×10^{-4}	0.0017	1.5×10^{-6}	0.35
Constant sample kernel	1.3	0.012	0.0022	0.007	0.48
Random neighbors	1.1	1.3	0.1	0.97	1
Noisy positive	1.4	4.4×10^{-4}	1.2×10^{-5}	0	0.36
Affinity Drifting	0.47	0.97	0.74	0.94	1
Oracle (16 samples)	0.5	0.91	0.83	0.78	0.98

Table 1. **2-D toy, held-out conditions** (mean over five seeds). Above the line: the one-positive objectives and three ablations of Affinity Drifting; below: Affinity Drifting and an oracle that sees 16 true samples per condition and step.

3.2. Mean collapse and its repair in a 2-D toy

Setup. Each of $N = 10{,}000$ training conditions $c \in [0, 1]^2$ has one sample from $p_{\text{data}}(\cdot \mid c)$, a three-component Gaussian mixture whose components rotate with c_1 and whose radius and weights change with c_2, so the conditional mean lies between modes. An MLP generator $f_\theta(\epsilon, c)$ sees c quantized to cells of width $\delta = 1/16$, about 39 training conditions per cell. We train with Adam (learning rate 10^{-3}, selected on the unconditional model, whose marginal Wasserstein-2 distance is 0.136 [0.129, 0.144] against a finite-sample floor of 0.0876), an exponential moving average of the weights, 20,000 steps and 8 conditions × 64 samples per batch, and evaluate 512 samples on each of 256 held-out conditions. We report the Wasserstein-2 distance W_2 to the true conditional, the ratio of total variances $\operatorname{tr}\Sigma_q / \operatorname{tr}\Sigma_p$ (1 for a correct spread, 0 for a point mass), precision and recall [5], the fraction of modes that receive at least 5% of the samples, and the fraction of samples below the 1% quantile of the true log-density (off-support samples).

One positive collapses. Both the original Drifting objective and the one-positive objective ($h \to 0$) collapse every condition to a point at the conditional mean (variance ratio 2.8×10^{-5} [$2.0 \times 10^{-5}, 3.4 \times 10^{-5}$] and 3.8×10^{-4} [$2.9 \times 10^{-4}, 4.7 \times 10^{-4}$]; off-support share 1), while the oracle, given 16 true samples per condition and step, covers the modes (variance ratio 0.909 [0.906, 0.915], modes captured 0.977; PASS).

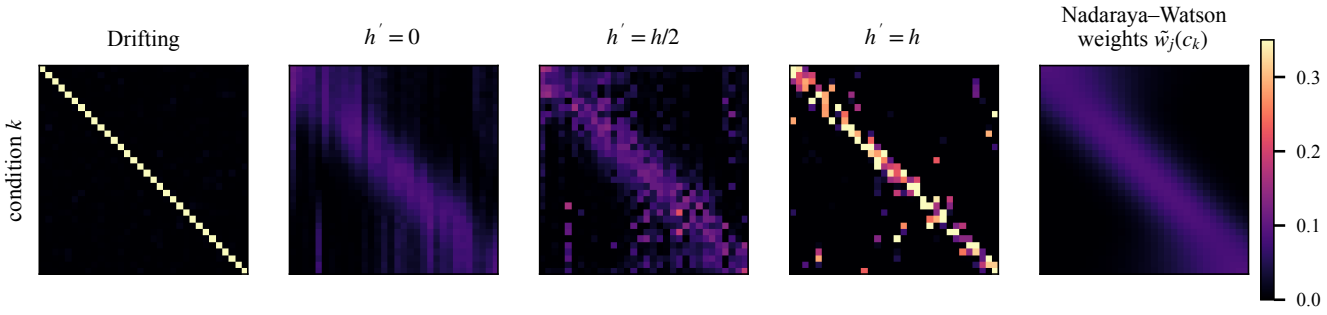


Figure 2. **The negative bandwidth h' selects the equilibrium** (particle system, mean over seeds). Entry (k, j): share of condition k's particles nearest to positive y_j. The original Drifting objective memorizes (diagonal). With $h' = 0$, mass spreads over the pooled neighbors as the Nadaraya–Watson weights $\tilde{w}_j(c_k)$ predict (right); larger h' narrows the band toward memorization.

Pooling positives restores the mixture. Trained on the same single positives, Affinity Drifting captures every mode (fraction 1, recall 0.94) and reaches a held-out W_2 of 0.469 [0.46, 0.477] against 1.41 [1.41, 1.42] for the one-positive objective (ratio 0.33; Tab. 1), on par with the oracle. The pre-registered comparison is PASS: Affinity Drifting improves on the one-positive objective and on every ablation, with paired confidence intervals that exclude zero. The ablations fail in the ways the mechanism predicts. With a constant sample-space kernel, the same pooled positives give an x-independent target, their weighted average, and the generator collapses as with one positive (variance ratio 0.0122). Pooling as many random neighbors spreads samples over all modes (fraction 1) but places most of them off the conditional support (off-support share 0.922). Perturbing the single positive with noise of matched variance still collapses (variance ratio 4.4×10^{-4}). The gain therefore comes from the x-dependent assignment of samples to the modes of semantically similar positives, not from more or smoother targets. Figure 3 shows this assignment: with one positive every attraction vector points at the same target, whereas the pooled positives partition the plane into one basin per mode. One caveat is already visible: with ample capacity, some samples land on pooled training positives (share within half the median nearest-neighbor spacing of a training positive: 0.165 on training conditions, against 0.00443 with one positive), as the fixed-point analysis allows.

Bandwidth. Scaling the bandwidth $h(c)$ selected by $n_{\text{eff}} = n^\star$ by a factor $m \in \{\frac{1}{4}, \frac{1}{2}, 1, 2, 4, 8\}$ gives a U-shaped held-out W_2 with its minimum at $m = 1$ (PASS): 0.925 at $m = \frac{1}{4}$, 0.469 at $m = 1$ and 0.77 at $m = 8$. Too small a bandwidth pools too few positives to resolve the modes; too large a bandwidth pools conditions whose modes lie elsewhere. A single bandwidth shared by all conditions, set to the median of the selected $h(c)$, performs alike in this toy (0.467), where conditions are uniformly dense. The unnormalized drift energy $\mathbb{E}\|V_{p,q}\|^2$ after training decreases monotonically in h (0.25 at $m = 1$, 0.125 at $m = 8$): a larger bandwidth gives a smoother target that is easier to match, not a better model. The drift energy is therefore not a criterion for selecting h; we never use it for selection.

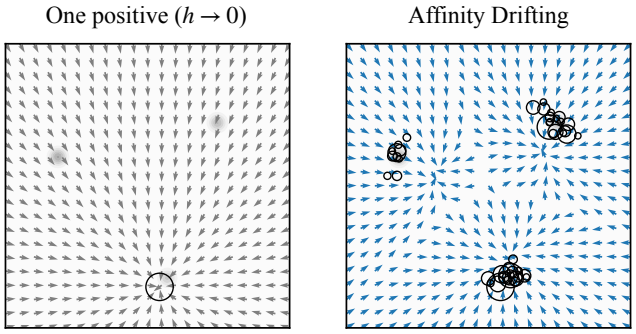


Figure 3. **Attraction field.** Direction of $V_p^+(x)$ for one training condition c. One positive ($p = \delta_y$): every vector points at the single target. Affinity Drifting ($p = \hat{p}_h(\cdot \mid c)$): vectors point to the nearest mode of the pooled positives (circles, area $\propto \tilde{w}_i(c)$).

The gap grows as conditional resolution decreases. [pending] (Fig. 4).

4. Status, deviations and limitations

This is a working draft; Tab. 2 lists every pre-registered experiment and its outcome at the time of the build. Changes to the plan are logged with reasons and approvals before any affected run; those that matter for reading the results are the following. The learning-rate selection failed twice before passing: first because the unconditional model was misspecified (negatives from each condition's samples rather than from the whole batch), then narrowly. A single kernel scale transported mass too locally and was replaced by three scales. 64 instead of 8 samples per condition are generated, since negatives estimated from 8 samples are strongly biased. The drift normalization was changed from separate



Figure 4. **Conditional resolution.** W_2 of the one-positive objective and of Affinity Drifting as the quantization of c seen by the generator coarsens (mean over seeds, 95% confidence interval).

normalization of V_p^+ and V_q^- to normalization of the whole field, as in [2], after the separate version sent one-positive samples outside the data. Runs made before a correction are kept and excluded from all results.

The toy uses the true condition coordinates as the embedding ψ; text embeddings, learned features and real captions are later stages of the study and are not reported here.

Setting	Pre-registered claim	Outcome
Particles	Mean shift equals the scaled score difference	PASS
Particles	Negatives with $h' = h$ memorize the positives	INCONCLUSIVE
Particles	Negatives with $h' = 0$: Nadaraya–Watson equilibrium	FAIL
Particles	Equilibrium moves monotonically with h'	PASS
Particles	Cross-condition negatives: closed-form equilibrium	FAIL
Particles	Same, under a Gaussian moment closure	PASS
Particles	Joint vs. separate softmax: same equilibrium	FAIL
2-D toy	Unconditional model fits the marginal (learning rate)	FAIL / FAIL / PASS
2-D toy	One positive, no negatives: ℓ_2 regression	PASS
2-D toy	One positive regresses to the conditional mean	PASS
2-D toy	Affinity Drifting restores the mixture; beats ablations	PASS
2-D toy	Gap grows as resolution in c coarsens (quantization)	[pending]
2-D toy	Same, with a Lipschitz constraint on c	[pending]
2-D toy	h' selects the equilibrium of a trained generator	[pending]
2-D toy	Bandwidth: U-shaped error, minimum near $n^\star$	PASS

Table 2. Pre-registered experiments and their outcomes at build time.

5. Related work

Affinity Drifting builds on drifting models [2], which turn kernel mean-shift [1] into a training signal for one-step generators. Its positive distribution is a Nadaraya–Watson kernel estimate [8, 10] of the conditional distribution, with the bandwidth set by the effective sample size of the weights [4]. The field is closely related to gradients of kernel two-sample statistics (MMD) [3], which generative models have combined with mixtures of kernel scales for the same reason we do [6]. The evaluation of mode collapse follows standard practice: precision and recall [5] and density and coverage [9]. Capacity is controlled by quantizing the condition and by spectral normalization [7].

References

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